

Writing Two-Step Equations from Real-World Contexts

Teacher's Edition

Standard and Targets

MA.7.AR.2.2 Write and solve two-step equations in one variable within a mathematical or real-world context, where all terms are rational numbers.

Boundary Forms are limited to $px \pm q = r$ and $p(x \pm q) = r$. Lessons 3 through 5 solved equations of those forms. This lesson is the first built entirely on writing them; Lesson 5 asks for one in its Example 3, its worksheet question 8 and its Ticket.

Learning targets I can write two-step equations to represent real-world contexts. I can solve two-step equations within a real-world context. I can say what the solution means in the situation.

Essential questions How do you write a two-step equation to represent a real-world context? How do you interpret the solution for a real-world context?

Building on MA.6.AR.2.2 and MA.6.AR.2.3 — one-step equations from context with integer terms.

Working toward MA.8.AR.2.1 — multi-step linear equations with rational coefficients, including variables on both sides.

Where this sits in the course Fourth consecutive lesson on MA.7.AR.2.2 and the sixth grade 7 lesson in a row to open Unit 1 — all of it required content these students are seeing for the first time. Writing the equation, rather than being handed one, is the part that transfers: **every context item on the Grade 8 FAST begins with this step**, whatever code it is filed under. Worksheet questions 9 and 10 and Additional Practice questions 9 and 10 are written at MA.8.AR.2.1, with the letter on both sides, so students meet the grade 8 form while the grade 7 habit is still being built.

Materials Whiteboards and markers. Calculators — this lesson has $12,340 - 780t = 5,125$ and $600 + 0.04s = 982$ kinds of arithmetic in it, and Math Nation says plainly that calculators keep the focus on reading the contexts. Worksheet 7A.01.06.

Lesson at a Glance

Windy Hill periods run **53 minutes**.

Segment	Slides	Time
Settle in; post the learning target	1	1 min
Warm-Up: same context, different question	2	4 min
Notes I–IV: define the letter, p is a rate and q happens once, percentages, 20% off	3–6	13 min
Example 1: a tower draining	7–9	5 min
Example 2: a commission	10–12	5 min
Example 3: the question asks for one of three numbers	13–15	5 min
Whiteboards: 9 questions, each with its reveal	16–33	15 min

Segment	Slides	Time
IXL — start today's skills	34	5 min
Total		53 min

Where the lesson lives: Notes II (slide 4) and Example 3 (slides 13–15). Notes II is where each number in the story finds its place in $px + q = r$, which is the skill the lesson is named after. Example 3 is the reading error, and it is the one students actually lose points to.

Read this before you plan the day. Math Nation's components total 45 minutes — 5 for the Warm-Up, 15 for Guided Instruction, 10 for the Try It, 10 for Guided Practice and 5 for the Wrap-Up — before any worksheet exists. Add a worksheet and it does not fit the period. The plan above fits because the Try It, which is a single four-part question, folds into Example 1, and the Guided Practice's two percentage problems become Notes III and IV.

No partner or group work anywhere in this lesson. Math Nation has three: Guided Instruction question 1b tells students to discuss the relationship with a partner, the Try It carries a tip about partners checking each other's work, and the literacy routine is Take Turns. Question 1b is now part of Example 1, which the whole class works together, and the Try It is Example 1 itself.

Vocabulary

commission — a percent of sales, paid on top of a salary

discount — a percent taken off a price

variable — a letter standing for the quantity that changes

constant — the quantity that stays the same no matter what

A note on the glossary. Math Nation lists “variables” as the only key term and adds commission and discount. Constant is added here because Notes II is built on the contrast between the rate and the starting value, and students need a name for the second one.

Slide-by-Slide Teaching Notes

Slide 1. Writing Two-Step Equations from Contexts *1 min*

- Post the learning target and name the shift honestly: the solving is Lesson 3 and they have it.
- Everything difficult about today happens before the first line of algebra.
- That framing stops the strong solvers from switching off during the notes.

Slide 2. Warm-Up *4 min*

- TYPE: SSDD.
- INDEPENDENT.
- ANSWERS: 1) $2000 - 60(5) = 1,700$ gallons, arithmetic only. 2) $2000 - 60t = 1400$, so $t = 10$ hours. 3) t = the number of hours the tank has been draining — a number, not a unit. Question 1 needs no equation; question 2 does. Question 3 is Notes I arriving before it is taught, and the answers will be poor today, which is the point.

- OFF THE SLIDE, YOURS TO SAY: The rate takes water away, so its sign is negative. That is question 2's whole difficulty.

Slide 3. Notes: I. Define the letter in a full sentence 3 min

- THE HABIT THIS LESSON IS FOR.
- Item C is the correction to make relentlessly: 't = hours' says t is a unit of time, which makes 60t nonsense.
- Write both versions on the board and ask what $60 \times \text{hours}$ would mean.
- OFF THE SLIDE, YOURS TO SAY: If you cannot finish the sentence, you cannot write the equation.

Slide 4. Notes: II. In $px + q = r$, p is a rate and q happens once 4 min

- The panel is the test that sorts p from q without keyword-hunting.
- If the answer doubles when the hours double, it is attached to the variable.
- Keywords fail on this benchmark often enough that a structural test is worth more.
- OFF THE SLIDE, YOURS TO SAY: Ask which number changes when the unknown changes. That number is p.

Slide 5. Notes: III. A percentage becomes a decimal first 3 min

- Students copy all three.
- The conversion has to happen before the equation is written, not during the solving — a percent sign left in an equation is the start of most of the wrong answers on this benchmark.
- OFF THE SLIDE, YOURS TO SAY: The decimal is the multiplier. The percent sign never appears in the equation.

Slide 6. Notes: IV. "20% off" means you pay 80% 3 min

- THE NAMED DIFFICULTY on this benchmark.
- Both 0.20p and 0.80p are correct answers to different questions, which is exactly why students choose wrongly — neither looks wrong.
- The panel is the deciding question and it is worth writing on the board for the week.
- OFF THE SLIDE, YOURS TO SAY: Read what the question asks for: the discount, or the price after it?

Slide 7. Example 1 1 of the 5 min

- SPOKEN ONLY. t = the number of hours the tower has been draining.
- Hold out for the full sentence.
- Then ask which number is p and which is q, using the panel test from Notes II.

Slide 8. Example 1 2 of the 5 min

- The minus in front of 780t is Notes II item D.
- Watch the second line: subtracting 12,340 from both sides gives a negative on both, and students who write 7,215 there get $t = -9.25$ and rarely notice.

Slide 9. Example 1 *2 of the 5 min*

- The second half of the panel is worth the ten seconds — 9.25 hours read as 9 hours 25 minutes is a common and invisible error, and it is the kind of thing the interpretation step exists to catch.
- OFF THE SLIDE, YOURS TO SAY: A positive number of hours, so the situation allows it. 0.25 of an hour is 15 minutes, not 25.

Slide 10. Example 2 *1 of the 5 min*

- SPOKEN ONLY.
- The 4% is p, because the earnings change when the sales change; the \$600 is q, paid whatever happens.
- And 4% becomes 0.04 BEFORE the equation is written — Notes III.

Slide 11. Example 2 *2 of the 5 min*

- Dividing by 0.04 is the step students dislike; remind them it is the same as multiplying by 25, or dividing by 4 and multiplying by 100.
- The check confirms the size of the answer, which looks surprising until it is verified.

Slide 12. Example 2 *2 of the 5 min*

- Notes IV as a table.
- The bottom row is the one that surprises them: 0.04p is a correct expression in both columns and answers a different question in each.
- Read each cell aloud as a sentence before moving on.
- OFF THE SLIDE, YOURS TO SAY: The same 0.04p is the extra in one column and the discount in the other.

Slide 13. Example 3 *2 of the 5 min*

- Whiteboards, equation and value only.
- The 10% becomes 0.10 before the equation is written — Notes III.
- Hold the class at the value; the question itself is the next slide and it is the part they lose points on.
- OFF THE SLIDE, YOURS TO SAY: Write the equation and solve it. Do not answer the question yet.

Slide 14. Example 3 *2 of the 5 min*

- Ask the question three ways in a row and make them answer each one: how much must she SELL — 4,500.
- What is her COMMISSION — 450.
- What does she EARN — 800.
- Then note that \$800 is not her commission even though it is the number the problem handed them; it is salary plus commission.

Slide 15. Example 3 *1 of the 5 min*

- Worksheet question 8 is this problem, and its answer key names all three quantities so you can mark the reading rather than the arithmetic.

- If a student hands back 450 or 800, they have done the mathematics and answered a different question — say exactly that, because it is a kinder and more accurate thing to hear than “wrong”.
- OFF THE SLIDE, YOURS TO SAY: That is why every problem today ends in a sentence, not a number.

Slide 16. Whiteboards · Question 1 of 9 — ask *about 20 seconds of thinking*

- Boards down until the cue, all up together, scan the back row first, and question a blank board before a wrong one.
- Do not reveal until every board is up.

Slide 17. Whiteboards · Question 1 of 9 — reveal *about 1 minute with the discussion*

- The baseline: an expression, not an equation.
- There is no equal sign because no total was given.
- Everything after this changes exactly one thing.
- COLOR: none on this sequence. One expression and its answer is not two objects and not two slots in a formula, so under rule 5 there is no axis to name and the answer color does the whole job.

Slide 18. Whiteboards · Question 2 of 9 — ask *about 20 seconds of thinking*

- Boards down until the cue, all up together, scan the back row first, and question a blank board before a wrong one.
- Do not reveal until every board is up.

Slide 19. Whiteboards · Question 2 of 9 — reveal *about 1 minute with the discussion*

- One feature changed: a total appeared.
- Notes II item C — r is the number the problem reports at the end, and it is the only thing that turns an expression into an equation.
- COLOR: none on this sequence. One expression and its answer is not two objects and not two slots in a formula, so under rule 5 there is no axis to name and the answer color does the whole job.

Slide 20. Whiteboards · Question 3 of 9 — ask *about 20 seconds of thinking*

- Boards down until the cue, all up together, scan the back row first, and question a blank board before a wrong one.
- Do not reveal until every board is up.

Slide 21. Whiteboards · Question 3 of 9 — reveal *about 1 minute with the discussion*

- One feature changed: now solve it.
- This is Lesson 3 and it should take fifteen seconds.
- Insist on the unit — 11 months, not 11.
- COLOR: none on this sequence. One expression and its answer is not two objects and not two slots in a formula, so under rule 5 there is no axis to name and the answer color does the whole job.

Slide 22. Whiteboards · Question 4 of 9 — ask *about 20 seconds of thinking*

- Boards down until the cue, all up together, scan the back row first, and question a blank board before a wrong one.
- Do not reveal until every board is up.

Slide 23. Whiteboards · Question 4 of 9 — reveal *about 1 minute with the discussion*

- One feature changed: q is gone.
- One step instead of two, and the answer stops being whole.
- Ask what 15.17 months means for a monthly membership — you would have paid for 16, which is Lesson 7's rounding question arriving early.
- COLOR: none on this sequence. One expression and its answer is not two objects and not two slots in a formula, so under rule 5 there is no axis to name and the answer color does the whole job.

Slide 24. Whiteboards · Question 5 of 9 — ask *about 20 seconds of thinking*

- Boards down until the cue, all up together, scan the back row first, and question a blank board before a wrong one.
- Do not reveal until every board is up.

Slide 25. Whiteboards · Question 5 of 9 — reveal *about 1 minute with the discussion*

- One feature changed: the rate is a percentage rather than a flat amount.
- Notes III and IV together.
- A board saying 0.15s has written the FEE, which is a correct expression for a different question.
- COLOR: none on this sequence. One expression and its answer is not two objects and not two slots in a formula, so under rule 5 there is no axis to name and the answer color does the whole job.

Slide 26. Whiteboards · Question 6 of 9 — ask *about 20 seconds of thinking*

- Boards down until the cue, all up together, scan the back row first, and question a blank board before a wrong one.
- Do not reveal until every board is up.

Slide 27. Whiteboards · Question 6 of 9 — reveal *about 1 minute with the discussion*

- THE ONE THAT CLOSES IT.
- Two features changed from question 5: added becomes off, and 15% becomes 25%. 1.15 became 0.75, and the structure is identical — you always multiply by what you END UP with.
- Put questions 5 and 6 side by side and ask what 0.25p would be the answer to.
- The discount itself, which nobody asked for.
- COLOR: none on this sequence. One expression and its answer is not two objects and not two slots in a formula, so under rule 5 there is no axis to name and the answer color does the whole job.

Slide 28. Whiteboards · Question 7 of 9 — ask *about 20 seconds of thinking*

- Boards down until the cue, all up together, scan the back row first, and question a blank board before a wrong one.

- Do not reveal until every board is up.

Slide 29. Whiteboards · Question 7 of 9 — reveal *about 1 minute with the discussion*

- One feature changed from the percentage pair: back to a flat rate, but now every number is a decimal and the growth is small. $6 + 0.08d = 6.48$.
- THE BOARD TO LOOK FOR SAYS 81, from dividing 6.48 by 0.08 without subtracting the 6 first — the same error as the plumber in Lesson 5, one lesson later, and it is worth saying that out loud.
- The 6 is what the giraffe already was; only 0.48 of the height was grown.
- Insist on the unit: 6 days, not 6.
- COLOR: none on this sequence. One expression and its answer is not two objects and not two slots in a formula, so under rule 5 there is no axis to name and the answer color does the whole job.

Slide 30. Whiteboards · Question 8 of 9 — ask *about 20 seconds of thinking*

- Boards down until the cue, all up together, scan the back row first, and question a blank board before a wrong one.
- Do not reveal until every board is up.

Slide 31. Whiteboards · Question 8 of 9 — reveal *about 1 minute with the discussion*

- One feature changed: the constant is not given to you — four dimes has to become \$0.40 before the equation can be written.
- THE BOARD TO LOOK FOR HAS $0.25q + 4 = 2.40$, which counts the dimes instead of valuing them and gives a negative number of quarters.
- That negative is a gift: Lesson 4 said a negative answer is wrong only against what the letter stands for, and you cannot have -6.4 quarters.
- The check is the sentence, not the arithmetic — read the answer back as money and it either sounds like \$2.40 or it does not.
- COLOR: none on this sequence. One expression and its answer is not two objects and not two slots in a formula, so under rule 5 there is no axis to name and the answer color does the whole job.

Slide 32. Whiteboards · Question 9 of 9 — ask *about 90 seconds of writing*

- Boards down until the cue, all up together, scan the back row first, and question a blank board before a wrong one.
- Do not reveal until every board is up.

Slide 33. Whiteboards · Question 9 of 9 — reveal *about 2 minutes with the closing discussion*

- WRITTEN WORK, and the only question in the round that asks for it.
- With the worksheet gone this is the only place in the period the habit gets built, and IXL takes an answer without ever seeing the work.
- ANSWER: $0.80p - 10 = 36.20$, so $0.80p = 46.20$ and $p = \$57.75$.
- Check it forward: 20% off 57.75 is 46.20, less the coupon is 36.20.
- THE ORDER IS THE WHOLE QUESTION and it is testable rather than arguable: doing the coupon first gives $(p - 10)(0.80) = 36.20$ and $p = \$55.25$, which is a DIFFERENT price.

- Both orders check out against their own equation, so the check cannot save you here — only re-reading the sentence can.
- That is the one thing in this lesson a calculator is no help with at all, and it is why question 7 of the bank is written the way it is.
- COLOR: none on this sequence. One expression and its answer is not two objects and not two slots in a formula, so under rule 5 there is no axis to name and the answer color does the whole job.

Slide 34. IXL 5 min

- IXL.
- The skills are assigned in Google Classroom, not named here, and they are deliberately not ours -- Croix: "The students need questions that aren't exclusively mine."
- Makes them think." Circulate.
- The thing to watch for is a student answering in the box with no paper in front of them: IXL takes an answer and never sees the work, so this is the one part of the period where the habit can quietly lapse.

Worksheet 7A.01.06 — Answer Key

1. water tower	$110,000 - 2,150h = 79,900$ Known: 110,000 gallons at the start; 2,150 gallons drained each hour; 79,900 gallons now. Unknown: h = the number of hours the tower has been draining. Do not solve. (For reference: $h = 14$.)
2. giraffe	$6 + 0.08d = 6.48$ Known: 6 feet at birth; 0.08 foot gained each day; 6.48 feet now. Unknown: d = the number of days old. Do not solve. (For reference: $d = 6$.)
3. toddler	$35.45 + 0.2m = 37$, so $m = 7.75$ It will take James 7.75 months. 0.2 is the rate and 35.45 is the starting value; watch for students who divide 37 by 0.2 and never subtract.
4. quarters and dimes	$0.25q + 0.40 = 2.40$, so $q = 8$ Mr. Humphries has 8 quarters. $25q + 40 = 240$ is equally correct — everything in cents. What is not correct is $25q + 40 = 2.40$, which mixes dollars and cents.
5. phone plan	$24.99 + 0.12g = 32.19$, so $g = 60$ 60 gigabytes. The \$24.99 is charged whether any data is used or not, so it is the constant.
6. concert tickets	$4(t + 2.50) = 118$, so $t = \\$27.00$ The fee is charged on each of the four tickets, so the group repeats and needs parentheses. $4t + 2.50 = 118$ charges the fee once and gives \$28.875.
7. Maddox, shoes	$0.80p - 10 = 36.20$, so $p = \\$57.75$ Taking 20% off means paying 80% — multiply by 0.80, not 0.20. The sale price is \$46.20, which is 80% of the original. The \$36.20 he actually paid is about 63% of the original, because the coupon comes off after the percentage.

8. Victoria, commission	$350 + 0.10s = 800$, so $s = \\$4,500$ Three quantities and the question asks for the first: sales \$4,500, commission \$450, total earnings \$800. A student who answers \$450 has solved correctly and reported the wrong number.
9. two gyms (grade 8)	$30 + 18m = 75 + 12m$, so $m = 7.5$ Collect the m terms first: $18m - 12m = 6m$, so $6m = 45$. Both gyms cost \$165 at 7.5 months.
10. two data plans (grade 8)	$45 + 0.10g = 30 + 0.25g$, so $g = 100$ Moving the smaller g term keeps the coefficient positive: $15 = 0.15g$. Both plans cost \$55 at 100 GB. Below that Plan B is cheaper; above it, Plan A.

Question 7 is the one Math Nation gets ambiguous. Their version says only that Maddox has a \$10 coupon and the shoes are 20% off, and never says which comes first. Both readings give clean answers — $0.80p - 10 = 36.20$ gives \$57.75, and $0.80(p - 10) = 36.20$ gives \$55.25 — so a student cannot rule either out. Ours states the order. If a student still writes the other equation, they have read carefully and made an arithmetic choice, not an error; show them both.

Question 8 is the reading question. Sales \$4,500, commission \$450, total earnings \$800. Math Nation's own answer key calls the \$800 a commission, which it is not.

Questions 9 and 10 are grade 8 and are not expected to be fluent yet. The equation is the part to look for; the solve may still stall and that is expected. What matters today is seeing that a letter can appear on both sides and that the same properties of equality still apply. Question 9 gives 7.5 months, which is a fine answer — memberships are billed monthly, but the question asks when the costs are equal, not when you would switch.

Additional Practice 7A.01.06 — Answer Key

Post this in Google Classroom alongside the practice sheet. It is optional work for students who want more.

1. realtor	$30,000 + 0.03s = 60,000$, so $s = \\$1,000,000$ Her salary is \$30,000 and her commission is \$30,000. The \$60,000 is the two added together.
2. Fiona, meal	$0.90p - 3 = 8.25$, so $p = \\$12.50$ 10% off means paying 90%. Multiply by 0.90, not 0.10.
3. storage tank	$4,500 - 125h = 2,750$, so $h = 14$ The rate is negative because the amount is going down.
4. eight campers	$8(4.25 + n) = 102$, so $n = \\$8.50$ Dividing both sides by 8 first is quicker here than distributing.
5. five books	$5(b + 1.60) = 73$, so $b = \\$13.00$ $5b + 1.60 = 73$ charges shipping once instead of five times and gives \$14.28 — close enough to look plausible, which is why the parentheses matter.
6. jacket	$0.70p - 5 = 47.50$, so $p = \\$75.00$ 30% off means paying 70%.
7. Devon, commission	$480 + 0.06s = 792$, so $s = \\$5,200$ His commission was \$312, not \$792. The \$792 is his total earnings.
8. Devin's error	<i>He mixed units: coins in cents, total in dollars. Correct: $0.25x + 0.40 = 2.40$ or $25x + 40 = 240$. Either gives $x = 8$.</i> These are the only two correct equations, so this is not an “answers will vary” question. A student who writes $0.25x + 40 = 2.40$ has made the same error in the other direction.
9. two movers (grade 8)	$90 + 40h = 150 + 28h$, so $h = 5$ Both companies charge \$290 at 5 hours. Shorter jobs favor the first; longer jobs favor the second.
10. Sara and Theo (grade 8)	$260 + 15w = 500 - 25w$, so $w = 6$ Adding $25w$ to both sides is what turns $-25w$ into a positive term. Both have \$350 after 6 weeks.

Misconceptions to Watch

- **Struggling to define the unknown.** Named first in Math Nation's list. Notes I and the whiteboard sequence. (Math Nation TE)
- **Misplacing the decimal when working with money and percentages.** Also named. Notes III and the calculator. (Math Nation TE)
- **Not seeing that p in $px + q = r$ is usually a rate and q is usually a starting value.** Also named. Notes II. (Math Nation TE)
- **Not seeing that p in $p(x + q) = r$ is usually the number of groups.** Also named. Worksheet question 6 and Additional Practice questions 4 and 5. (Math Nation TE)
- **Multiplying by the discount instead of by what is left.** $0.20p$ instead of $0.80p$. Notes IV items A and B and the whiteboard sequence.

- **Reporting the wrong one of several dollar amounts.** Answering \$450 or \$800 when the question asked for sales. Slides 13–15 and worksheet question 8.
- **Writing an equation with no time or count in it.** $12,340 - 780 = 5,125$. Example 1. (Florida B1G-M)
- **Mixing dollars and cents in one equation.** $25q + 40 = 2.40$. Worksheet question 4 and Additional Practice question 8.

Changes from the Math Nation Lesson

- Two answer keys were corrected. In 1.6.4 question 1d Math Nation writes “Victoria still needs to sell \$4,500 worth of product to earn a commission of \$800 this week.” Her commission is \$450; the \$800 is her total earnings. The word “still” is also unsupported — the problem gives her no prior sales. In 1.6.4 question 2b they ask what percentage of the original price Maddox PAID and key it 80%. He paid \$36.20 on a \$57.75 original, which is about 63%. The 80% is the sale price before the coupon.
- The Maddox and Fiona problems now state the order of the two discounts. As printed, neither says whether the percentage or the coupon comes off first, and both readings produce clean answers, so a student has no way to choose.
- Math Nation's warm-up was replaced. It prints an error-analysis item whose division has the bar under only two of the three terms and a floating minus sign, so it reads as a typesetting fault rather than as the student error it is meant to be. The new warm-up is an SSDD item on one context — a tank holding 2,000 gallons draining at 60 gallons an hour — asking an arithmetic question, an equation question, and for the definition of the letter.
- The Try It grew from one question to a lesson component. As printed it is a single four-part problem for a ten-minute block. It is now Example 1, across its three slides, plus the first four worksheet questions.
- The wrap-up's “answers will vary” was removed. Only two equations correctly represent Mr. Humphries's pocket — $0.25x + 0.40 = 2.40$ and $25x + 40 = 240$ — and the printed key invites students to think anything is acceptable. Additional Practice question 8 asks for both by name.
- Notation was cleaned throughout. The printed keys use “.2m” with no leading zero, and print $25x \div 25 = 200 \div 25$ as a stacked fraction whose bar spans the wrong terms.
- Every partner and group activity is replaced — three of them. Guided Instruction 1b becomes part of Example 1, worked as a class; the Try It partner-check tip is gone; Take Turns becomes the whiteboard rounds.
- The word “homework” was removed. Math Nation's tip reads “if time is an issue, consider assigning Try It for homework.” Worksheet 7A.01.06 is classwork; the Additional Practice goes to Google Classroom for students who want more.
- Worksheet questions 9 and 10 and Additional Practice questions 9 and 10 were added at MA.8.AR.2.1 — the unknown on both sides. Nothing in the grade 7 content was changed to make room; these are additions at the end of each sheet.
- The Wrap-Up was replaced with a Ticket Out the Door that produces gradeable evidence: a variable defined in a sentence, a discount written as a factor, and a commission equation solved and interpreted.
- Answer keys were built for both sheets from the same source file as the student versions, so the two cannot drift apart.

- Apart from the two corrected keys and the stated discount order, nothing in the mathematics was changed.